

Conv-ChArT — pose accuracy, four systems

Rotation in **degrees** (deg) — the geodesic angle between true and estimated board rotation. Translation in **board squares** (sq): the evaluation set fixes square_length_m = 1.0, so the unit is board squares, not metres. 1500 images.

All four algorithms share the identical downstream pipeline — lattice gate, recovery pass, then PnP (IPPE) — so the only thing that differs between rows is the corners each detector produced. Classical OpenCV and Deep ChArUco are therefore *not* using their own pose solvers; these numbers will not match their native output, by design. A **refusal** (a pose the pipeline declines to corroborate) is counted in the solve rate and excluded from the error statistics — it is a correct outcome, not an error.

algorithm	solve %	rot		rot p95 (deg)	trans		trans p95 (sq)
		median (deg)	rot mean (deg)		median (sq)	trans mean (sq)	
Conv-ChArT (coarse)	94.3	0.362	0.838	2.077	0.0272	0.0783	0.2788
Conv-ChArT + refiner	94.3	0.137	0.478	1.035	0.0104	0.0415	0.1545
Deep ChArUco (fine-tuned)	94.5	0.873	10.771	90.468	0.0775	1.2615	7.3668
classical OpenCV	54.9	0.148	0.991	2.209	0.0191	0.0720	0.2230
<i>lead: refined vs Deep ChArUco</i>	1.00x	6.4x	22.5x	87.4x	7.4x	30.4x	47.7x
<i>lead: refined vs classical</i>	1.7x	1.1x	2.1x	2.1x	1.8x	1.7x	1.4x

Reading the table

- **The two lead rows read as “how many times better our refined arm is”**, computed per column in that column’s own direction: solve % is higher-is-better, every error column is lower-is-better. A value below 1.0 means we are behind on that column.
- **The refiner’s contribution at pose level:** 2.6x on rotation median and 2.6x on translation median. Pose is what a docking controller consumes.
- **Classical OpenCV is accurate when it succeeds** — its median rotation is competitive with our coarse detector — but it only solves 55% of frames. The claim against it is coverage, not precision.
- **Deep ChArUco has a heavy tail.** Its median is respectable, but p95 rotation is 90 degrees: confidently wrong poses on several percent of frames. For a docking robot a wrong pose is more dangerous than no pose, which makes the p95 column more operationally important than the median.
- **Solve rate and p95 must be read together.** Solve rate alone flatters Deep ChArUco; p95 alone hides that classical refuses nearly half the time.